

Papathanasiou's dense generators are not C*-minimal

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Status

This is a claimed full negative answer, subject to human review, to the minimality question in Section 4 of Franca–Garces [1].

Source passage

The source paper constructs a copy $A_3 \cong \ell_\infty$ inside $(\ell_\infty \setminus c_0) \cup \{0\}$. In the final considerations, after constructing another subalgebra A_4 with a minimal generator system of cardinality $\mathfrak{c}$, it says:

Although it is even possible to find a dense subalgebra of A_3 as in [2], we do not know if that system of generators is minimal when generating ℓ_∞ (or A_3) as a C*-algebra.

The relevant source crop is included as Figure 1.

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of generators for $A := \text{Span}(\text{word}(S))$. Indeed, fix j and set $S_j = S \setminus \{p_j\}$. If $p_j = \sum_{k=1}^n \lambda_k a_k$ where each a_k is a finite product of elements in S_j , then $p_j = p_j^2 = \sum_{k=1}^n \lambda_k (p_j a_k)$. But is easy to realize that each of the elements $p_j a_k$ has finite support and hence $\sum_{k=1}^n p_j a_k$ also has finite support, a contradiction. The same argument (with $a_k \in S_j$) actually shows that the p_j 's are linearly independent and hence $\dim(A) = \mathfrak{c}$. Even though the algebra A is not contained in $(\ell_\infty \setminus c_0) \cup \{0\}$, we can find a subalgebra A_4 of A_3 with the same properties as A (without the need to perturbate the generators), since ℓ_∞ is *-isomorphic to A_3 . Although it is even possible to find a dense subalgebra of A_3 as in [22], we do not know if that system of generators is minimal when generating ℓ_∞ (or A_3) as a C*-algebra. We are going to sum up our findings in table [2] but before, we shall make an interesting last observation. Recall that if X is a separable Banach space, then it is possible construct an isometric copy X' of it in ℓ_∞ by [1]. As we already mentioned, if we directly apply [1], we cannot guarantee that $X' \cap c_0 = \{0\}$. However, once we have found a copy $A_3 \cong \ell_\infty$ with $A_3 \cap c_0 = \{0\}$ then a copy X'' of X in A_3 (and hence in ℓ_∞ itself) can be found such that $X'' \cap c_0 = \{0\}$.

Theorem 4.1. *There exists a *-isomorphic copy D of ℓ_∞ inside ℓ_∞ such that $D \cap c_0 = \{0\}$. Moreover, for every separable Banach space X there exists an isometric*

Figure 1: The source passage on page 17 of the rendered arXiv source.

Papathanasiou's family

In [2, proof of Theorem 2], for every $A \subseteq \mathbb{N}$ and $j \in \mathbb{N}$ there is a bounded sequence $f_{(A,j)}$ with $\|f_{(A,j)}\| \leq 1/j$, and

$$h_{(A,j)} = \chi_A + f_{(A,j)}.$$

Hence $h_{(A,j)} \rightarrow \chi_A$ in the sup norm as $j \rightarrow \infty$, for each fixed A . The generated algebra is dense in ℓ_∞ , and the proof in [2] shows that it lies in $(\ell_\infty \setminus c_0) \cup \{0\}$ and is generated by an algebraically independent family. The same argument is available over the complex scalars, as noted in [2, Remark 3].

Proof intuition

The density mechanism has one redundant coordinate built into it. For each set $A \subseteq \mathbb{N}$, the construction does not use one approximant $h_{(A,j)}$, but an entire tail converging to χ_A . Removing a single generator removes at most one term from one of these tails. All characteristic functions χ_A therefore remain in the C^* -closure of the surviving family, and those characteristic functions already generate ℓ_∞ as a C^* -algebra.

Theorem 1. *Let*

$$H = \{h_{(A,j)} : A \subseteq \mathbb{N}, j \in \mathbb{N}\} \subseteq \ell_\infty$$

be Papathanasiou's generator family. For every $(A_0, j_0) \in \mathcal{P}(\mathbb{N}) \times \mathbb{N}$,

$$C^*(H \setminus \{h_{(A_0, j_0)}\}) = \ell_\infty.$$

Consequently H is not a minimal C^ -generating set for ℓ_∞ . If $\Phi : \ell_\infty \rightarrow A_3$ is any $*$ -isomorphism onto the copy A_3 appearing in [1], then $\Phi(H)$ is not a minimal C^* -generating set for A_3 .*

Proof. Fix (A_0, j_0) , set

$$H_0 = H \setminus \{h_{(A_0, j_0)}\},$$

and let $B = C^*(H_0)$, the C^* -subalgebra of ℓ_∞ generated by H_0 .

We first show that B contains every characteristic function χ_A , $A \subseteq \mathbb{N}$. If $A \neq A_0$, then $h_{(A,j)} \in H_0$ for all j , and since $\|h_{(A,j)} - \chi_A\| \leq 1/j$, closedness of B gives $\chi_A \in B$. If $A = A_0$, choose any sequence $j_n \rightarrow \infty$ with $j_n \neq j_0$ for all n . Then $h_{(A_0, j_n)} \in H_0$ and

$$\|h_{(A_0, j_n)} - \chi_{A_0}\| \leq 1/j_n \rightarrow 0,$$

so again $\chi_{A_0} \in B$.

Thus B contains χ_A for every $A \subseteq \mathbb{N}$. Every finite-valued bounded sequence on $\mathbb{N}$ is a finite linear combination of such characteristic functions, so all finite-valued bounded sequences lie in B . Finally, finite-valued bounded sequences are uniformly dense in ℓ_∞ : given $x \in \ell_\infty$ and $\varepsilon > 0$, partition the bounded range of x into finitely many subsets of diameter less than ε , and approximate x by the corresponding step sequence. Since B is norm closed, $B = \ell_\infty$.

This proves that deleting the single generator $h_{(A_0, j_0)}$ still leaves a C^* -generating family for ℓ_∞ . As (A_0, j_0) was arbitrary, no element of H is indispensable, so H is not minimal.

If $\Phi : \ell_\infty \rightarrow A_3$ is a $*$ -isomorphism, then

$$C^*(\Phi(H) \setminus \{\Phi(h_{(A_0, j_0)})\}) = \Phi(C^*(H \setminus \{h_{(A_0, j_0)}\})) = \Phi(\ell_\infty) = A_3.$$

Hence the transferred system in A_3 is not minimal either. □

Verification notes

The proof uses only the displayed approximation $\|h_{(A,j)} - \chi_A\| \leq 1/j$, the fact that the index set contains all pairs (A,j) , and the elementary density of finite-valued sequences in ℓ_∞ . No set-theoretic assumptions are used. The argument is unchanged after passing through a *-isomorphism $\ell_\infty \rightarrow A_3$.

Novelty and search bounds

Before promotion, the lightweight run indexes were searched for arXiv id 2508.19174, the source title, “compact multipliers”, “algebrability”, “Papathanasiou”, “dense subalgebra”, and generator-minimality phrases. No existing packet or attempt answered this target. A bounded web search on 2026-07-01 for the exact source sentence, the source title, and close phrases combining Papathanasiou, $\ell_\infty \setminus c_0$, minimal generators, and C*-algebras did not surface an exact duplicate in returned snippets.

Human review recommendation

The only point requiring review is whether the phrase “that system of generators” in [1] refers exactly to the family $\{h_{(A,j)}\}$ from [2], or to its image under a fixed *-isomorphism $\ell_\infty \rightarrow A_3$. The theorem covers both readings.

References

- [1] W. Franca and Jorge J. Garces, *Applications of compact multipliers to algebrability of $(\ell_\infty \setminus c_0) \cup \{0\}$ and $(B(\ell_2(\mathbb{N})) \setminus K(\ell_2(\mathbb{N}))) \cup \{0\}$* , arXiv:2508.19174v1, 2025.
- [2] Dimitris Papathanasiou, *Dense lineability and algebrability of $\ell^\infty \setminus c_0$* , Proc. Amer. Math. Soc. 150 (2022), 991–996; arXiv:2102.03199.